

Class 9 Mathematics: Circles

The Ultimate Study Guide | CBSE Geometry Notes & Theorem Proofs

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Hello! Welcome to our comprehensive Class 9 Circles Master Session presented by **Chauhan Publishers**. Circles are one of the most elegant and symmetric shapes in geometry, appearing everywhere from nature to engineering. In this session, we will break down the entire Class 9 circles chapter from the basic vocabulary up to high-level proofs and NCERT exemplar problems. By the end of this guide, every concept in this chapter will be perfectly clear to you.

1. Foundations: Symmetries & Key Definitions

What is a Circle? In nature, circles form when raindrops fall on water, and they are reflected in the cross-sections of plant stems, the inflorescences of sunflowers, the full moon, and the sun during a solar eclipse. Mathematically, we define a circle based on this uniform, symmetrical roundness:

Mathematical Definition of a Circle

A circle is the set (or locus) of all points in a two-dimensional plane that lie at a constant, equal distance from a fixed point on that same plane.

The fixed point is called the **centre** of the circle, and the constant distance is called its **radius**.

Symmetries of a Circle

Circles are perfectly symmetrical shapes, which is why rotating wheels look identical at all times:

- **Rotational Symmetry:** A circle has complete rotational symmetry about its centre through any angle. No matter how much you spin it, its shape looks exactly the same.
- **Reflection Symmetry:** A circle has complete reflection symmetry. If you fold a paper circle so that the boundaries overlap, the resulting crease represents a line of reflection symmetry. All such creases pass through the centre and represent the circle's diameters.

Key Elements of a Circle

1. **Radius (r):** The line segment (or distance) from the centre of the circle to any point on its boundary.
2. **Chord:** A line segment joining any two distinct points on the circle's boundary.
3. **Diameter (d):** A chord that passes through the centre of the circle. It is the longest chord in the circle, and its length is equal to twice the radius ($d = 2r$). There is no chord longer than the diameter.
4. **Arc:** A connected portion of the circle's boundary. Two points A and B on a circle divide it into two arcs:
 - *Minor Arc:* The smaller portion of the boundary (subtending an angle $< 180^\circ$ at the centre).
 - *Major Arc:* The larger portion of the boundary (subtending an angle $> 180^\circ$ at the centre).
5. **Segment:** The region enclosed between a chord and its corresponding arc:
 - *Minor Segment:* Enclosed by the chord and the minor arc.
 - *Major Segment:* Enclosed by the chord and the major arc.

2. How Many Circles? (The 1, 2, and 3-Point Rules)

An essential question in geometry is: Given a set of points, how many circles can we draw through them?

The One-Point Rule: Through a single point A on a plane, you can draw infinitely many circles of any size.

The Two-Point Rule: Through two points A and B on a plane, you can draw infinitely many circles.

- **The Center Rule:** The centre of every circle passing through A and B must be equidistant from both points ($OA = OB$). Therefore, the centres of all these circles lie on the perpendicular bisector of the line segment AB.
- **Smallest Circle:** The smallest possible circle passing through A and B is the one where AB is the diameter. Its radius is exactly $\frac{1}{2}AB$, and its centre is the midpoint of AB.

The Three-Point Rule

Theorem: Circles Through Three Points

Statement: There is one and only one (unique) circle passing through three given non-collinear points.

Why is this true? If we have three points A, B, and C that are non-collinear (not in a straight line):

- 1. Any circle passing through A and B must have its centre on the perpendicular bisector of AB.
- 2. Any circle passing through B and C must have its centre on the perpendicular bisector of BC.
- 3. Since A, B, and C are non-collinear, these two perpendicular bisectors are not parallel and will intersect at a single, unique point O.
- 4. This intersection point O is equidistant from A, B, and C ($OA = OB = OC$). Using O as the centre and OA as the radius, we can draw exactly one circle that passes through all three points.

Circumcircle and Circumcentre: The unique circle passing through the three vertices of a triangle $\triangle ABC$ is called the circumcircle of the triangle, and its centre O is called the circumcentre.

Type of Triangle	Position of Circumcentre (O)	Visual Context
Acute-Angled	Inside the triangle	Balanced sides, perpendiculars meet inside.
Obtuse-Angled	Outside the triangle	Lies opposite the largest obtuse angle.
Right-Angled	At the midpoint of the hypotenuse	Hypotenuse acts as the diameter (90° angle).

What about Three Collinear Points?

No circle can pass through three collinear points. Because if A, B, and C lie on a straight line, the perpendicular bisectors of AB and BC are parallel to each other. They will never intersect, meaning there is no point that is equidistant from all three points!

3. CBSE Class 9 Circles Theorem Matrix

Here is a summary matrix of all core theorems for quick reference and revision:

#	CBSE Core Theorem Statement	Core Intuition (In Plain Words)
T1	Equal chords of a circle subtend equal angles at the centre.	Identical chords make identical triangular wedges at the center.
T2	If the angles subtended by chords at the centre are equal, then the chords are equal.	If the central "wedges" have the same angle, the chord lengths must match.
T3	The perpendicular from the centre of a circle to a chord bisects the chord.	A straight vertical line dropped from the center splits a chord in half.
T4	The line joining the centre and the midpoint of a chord is perpendicular to the chord.	Connecting the center to a chord's exact middle always creates a 90° angle.

#	CBSE Core Theorem Statement	Core Intuition (In Plain Words)
T5	Equal chords of a circle are equidistant from the centre.	Equal chords sit at the exact same depth from the circle's center.
T6	Chords equidistant from the centre of a circle are equal in length.	Chords at equal depths from the center are of equal length.
T7	Given two unequal chords, the longer chord is closer to the centre.	The longer a chord is, the deeper it goes into the circle (closer to the center).
T8	The angle subtended by an arc at the centre is double the angle subtended by it at the boundary.	The "opening" angle at the center is always twice the angle made at the circle's edge.
T9	Angles in the same segment of a circle are equal.	Any angles drawn from the same endpoints to the same segment are identical.
T10	The angle subtended by a diameter at any point on the circle is 90° .	Any triangle built on the diameter has a perfect 90° right angle at the boundary.
T11	The sum of either pair of opposite angles of a cyclic quadrilateral is 180° .	Opposite corners of a four-sided shape in a circle always add up to 180° .
T12	If opposite angles of a quadrilateral sum to 180° , the quadrilateral is cyclic.	If opposite corners add to 180° , you can draw a perfect circle through all four corners.

4. Deep-Dive Theorem Proofs & Key Definitions

Theorem T1: Central Angles of Equal Chords

Core Intuition: Think of a circle as a highly symmetrical wheel. Spinning the wheel doesn't deform it. If you have two chords of the exact same length, they form congruent triangular slices when connected to the center, meaning the angles at the center of those slices must be equal.

Statement: Equal chords of a circle subtend equal angles at the centre.

Given: A circle with centre C and two equal chords AB and DE (so $AB = DE$).

To Prove: $\angle ACB = \angle DCE$.

Step	Mathematical Statement	Geometric Reason
1	In triangles $\triangle CAB$ and $\triangle CDE$:	We compare the corresponding sides of both triangles.
2	$CA = CD$	Radii of the same circle (r).
3	$CB = CE$	Radii of the same circle (r).
4	$AB = DE$	Given (chords are of equal length).
5	$\triangle CAB \cong \triangle CDE$	By SSS (Side-Side-Side) Congruence Criterion.
6	$\angle ACB = \angle DCE$	By CPCT (Corresponding Parts of Congruent Triangles).

Hence Proved!

Theorem T3 & T4: Perpendicular from Center to Chord

Core Intuition: A triangle formed by the center and any chord is always isosceles (since the other two sides are equal radii). In any isosceles triangle, the perpendicular dropped from the vertex to the base always bisects the base.

Statement (T3): The perpendicular from the centre of a circle to a chord bisects the chord.

Given: A circle with centre O, chord AB, and line segment $OM \perp AB$ (so $\angle OMA = \angle OMB = 90^\circ$).

To Prove: $AM = BM$.

Step	Mathematical Statement	Geometric Reason
1	In right-angled triangles $\triangle OMA$ and $\triangle OMB$:	Comparing triangles on either side of OM.
2	$\angle OMA = \angle OMB = 90^\circ$	Given ($OM \perp AB$).
3	$OA = OB$	Radii of the same circle (Hypotenuse r).
4	$OM = OM$	Common side.
5	$\triangle OMA \cong \triangle OMB$	By RHS (Right-Hypotenuse-Side) Congruence.
6	$AM = BM$	By CPCT.

Hence Proved!

Theorem T8: The Double Angle Theorem

Statement: The angle subtended by an arc at the centre is double the angle subtended by it at any point on the remaining part of the circle.

Given: An arc AFB subtending angle $\angle ACB$ at the centre C, and angle $\angle ADB$ at point D on the boundary.

To Prove: $\angle ACB = 2 \angle ADB$.

Step	Mathematical Statement	Geometric Reason
1	Join D to centre C and extend it to E.	Construction to split our angles.
2	In $\triangle DCB$, $CB = CD$	Both equal to radius (r).
3	$\angle CBD = \angle CDB$	Angles opposite to equal sides.
4	$\angle BCE = \angle CBD + \angle CDB$	Exterior Angle Theorem ($\angle BCE$ is exterior).
5	$\angle BCE = 2 \angle CDB = 2 \angle BDC$	Substituting Step 3 into Step 4.
6	Similarly, in $\triangle DCA$: $\angle ACE = 2 \angle CDA$	Using same logic for left triangle.
7	$\angle ACB = \angle BCE + \angle ACE$	Angle Addition.
8	$\angle ACB = 2 \angle BDC + 2 \angle CDA = 2(\angle BDC + \angle CDA)$	Substituting Steps 5 and 6.
9	$\angle ACB = 2 \angle ADB$	Since $\angle ADB = \angle BDC + \angle CDA$.

Hence Proved!

5. Systematically Solved Practice Problems

Problem 1: Calculating Chord Length (Pythagorean Application)

Problem Statement: In a circle of radius 13 cm, a chord is drawn at a distance of 5 cm from the centre. Find the length of this chord.

Step-by-Step Solution:

1. Let the circle have its centre at O and radius $R = 13$ cm. Let the chord be AB.
2. The distance from the centre to the chord is perpendicular line segment $OM \perp AB$. Thus, $OM = 5$ cm.
3. Connect O to endpoint A. OA is a radius, so $OA = 13$ cm.

4. This forms right-angled triangle $\triangle OMA$ with $\angle OMA = 90^\circ$.

5. Apply Pythagoras theorem to $\triangle OMA$:

$$OA^2 = OM^2 + AM^2$$

$$13^2 = 5^2 + AM^2$$

$$169 = 25 + AM^2$$

$$AM^2 = 144 \Rightarrow AM = 12 \text{ cm}$$

6. By Theorem T3, perpendicular from centre bisects chord, so $AB = 2 \times AM = 2 \times 12 = 24 \text{ cm}$.

Answer: The length of the chord is **24 cm**.

Common Pitfall: The Half-Chord Trap

Many students solve for $AM = 12 \text{ cm}$ and write that as their final answer. Remember, AM is only half of the chord length! Always multiply by 2 to get the full chord length.

Problem 2: Finding a Nested Segment Length

Problem Statement: In a circle of radius 5 cm, a chord AB of length 8 cm is drawn. A line segment OD is drawn from the centre O perpendicular to AB, intersecting AB at C and the circle boundary at D. Find the length of CD.

Step-by-Step Solution:

1. Given radius $OA = 5 \text{ cm}$ and chord $AB = 8 \text{ cm}$.

2. Since $OD \perp AB$, segment $OC \perp AB$.

3. By Theorem T3, OC bisects AB. Therefore, $AC = BC = \frac{1}{2}AB = 4 \text{ cm}$.

4. In right-angled triangle $\triangle OCA$:

$$OA^2 = OC^2 + AC^2$$

$$5^2 = OC^2 + 4^2$$

$$25 = OC^2 + 16 \Rightarrow OC^2 = 9 \Rightarrow OC = 3 \text{ cm}$$

5. OD is a radius from centre O to boundary D, so $OD = 5 \text{ cm}$.

6. To find segment CD: $CD = OD - OC = 5 - 3 = 2 \text{ cm}$.

Answer: The length of segment CD is **2 cm**.

6. Quick-Recall Study Tips for CBSE Exams

Secret Key to Circles Chapter

Nearly 80% of all circle theorem proofs and numerical problems are solved using one simple trick:

Congruent Triangles ($\triangle \cong \triangle$) or the **Pythagoras Theorem** ($a^2 + b^2 = c^2$).

Exam Cheat-Sheet Checklist

- ✓ **Radii are always equal:** Whenever you see a circle problem with lines drawn to the boundary from the center, immediately label them as equal (r). It instantly reveals hidden isosceles triangles!
- ✓ **The Semicircle Rule:** Any angle sitting on the diameter is 90° . If you see a diameter, look for a right-angled triangle.
- ✓ **Opposite Angles in Cyclic Quadrilaterals:** If you are given three corners of a cyclic quadrilateral, the fourth is just 180° minus the opposite corner.
- ✓ **The Center Perpendicular:** Whenever a line is perpendicular from the center to a chord, draw a radius to the chord's endpoint to form a right-angled triangle.

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